

SHOCKLEY DIODE EQUATION

- The V-I relationship for a semiconductor diode is described by Shockley diode equation :

$$I = I_S \left(e^{\frac{qV}{\eta KT}} - 1 \right) = I_S \left(e^{\frac{V}{\eta V_T}} - 1 \right)$$

or
$$V = \eta V_T \ln \left(\frac{I}{I_S} + 1 \right)$$

- The physical constants fundamental to the diode V-I relationships are :

q = electronic charge = 1.6×10^{-19} Coulomb.

K = Boltzmann's constant = 1.38×10^{-23} Joule/°K

V_T = Voltage equivalent temperature of the diode.

= $KT/q \approx T/11600 \approx 26 \text{ mV @ room temperature (300°K)}$

- In strong forward bias $V \gg V_T$, the diode current expression becomes

$$I = I_S e^{\frac{V}{\eta V_T}}$$

- In strong reverse bias $V \ll -V_T$, the diode current expression becomes

$$I = -I_S$$

DYNAMIC RESISTANCE r_d

When the diode is in the strongly forward biased region the current experiences exponential growth and the diode appears to have non zero dynamic resistance.

The diode dynamic resistance is given by :

$$r_d = \frac{\partial V}{\partial I} = \frac{\eta V_T}{I_S} e^{-\frac{V}{\eta V_T}} = \frac{\eta V_T}{I_S + I} \approx \frac{\eta V_T}{I}$$

In strong forward bias $V \gg V_T$ or $I \gg I_S$

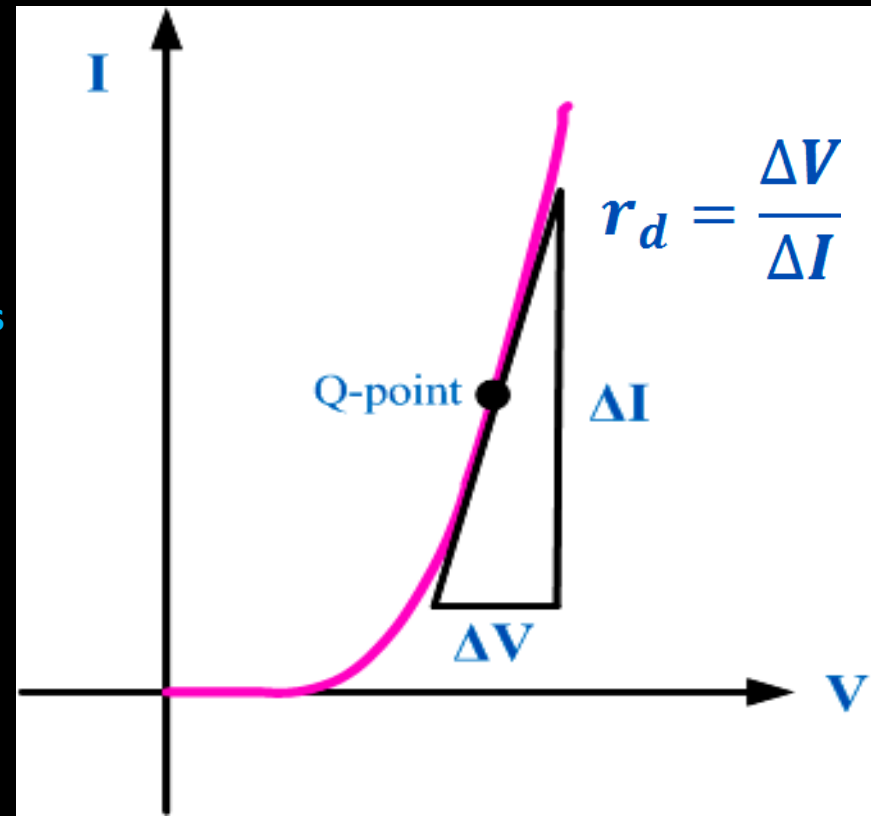
$$r_d \approx 0$$

In strong reverse bias $V \ll -V_T$ or $I = -I_S$

$$r_d \approx \alpha$$

Putting $\eta = 1$ and $V_T = 26 \text{ mV}$

$$r_d \approx \frac{26 \text{ mV}}{I_D}$$



Threshold or Cut-in Voltage V_γ

- ❑ The boundary between the conducting and the nonconducting region of **ideal diode** is at the origin of the V-I transfer characteristics.
- ❑ For **real diode** (pn-junction diode) the transition between the two regions lies at a positive diode voltage called the **threshold voltage or the cut-in voltage V_γ** .
- ❑ The **cut-in voltage V_γ** of the pn-junction diode is such that below which the current is very small (say less than one percent of maximum rated value).
- ❑ Beyond V_γ the current rises rapidly. Thus V_γ determines the boundary between the conducting and nonconducting regions of diode.
- ❑ The value of the threshold voltage V_γ can be derived by finding **the intersection of the tangent line at Q point to the voltage axis of the diode.**

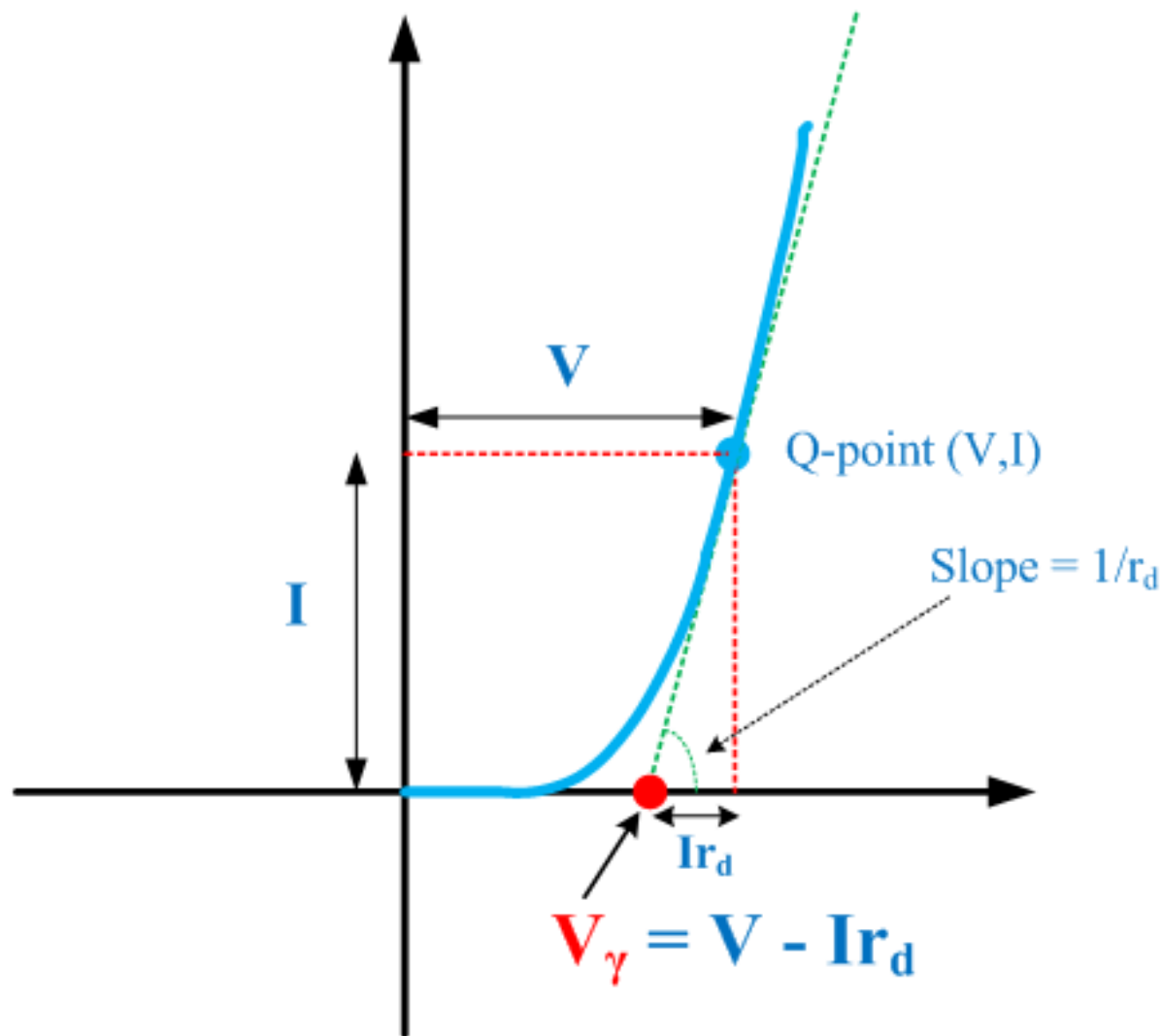
$$V_\gamma = V - Ir_d = V - \eta V_T \left(1 - e^{-\frac{V}{\eta V_T}} \right)$$

$V_\gamma(\text{Si}) = 0.6V \text{ to } 0.9V$

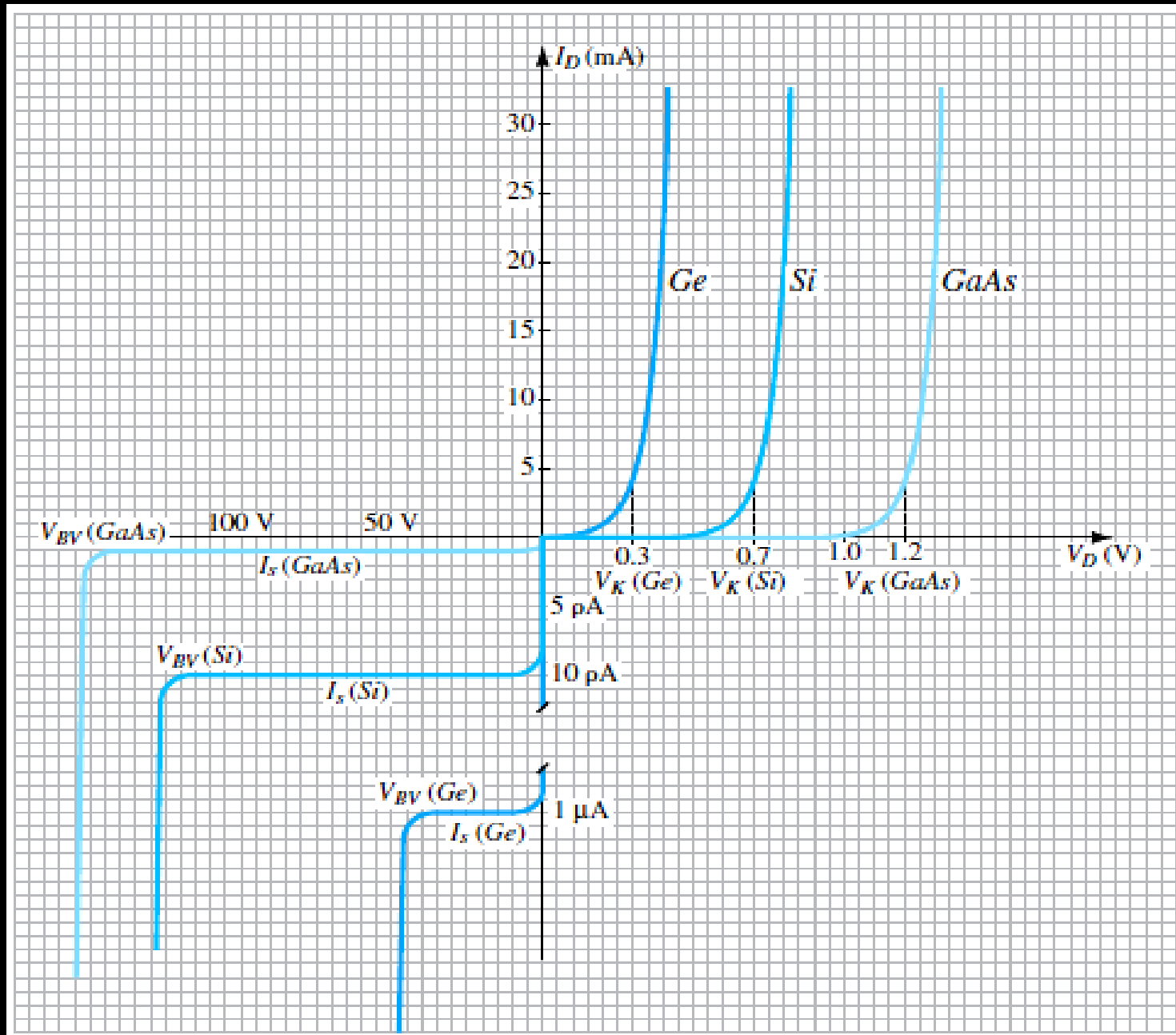
$V_\gamma(\text{Si})_{\text{typical}} = 0.7V$

$V_\gamma(\text{Ge}) = 0.3V$

$V_\gamma(\text{GaAs}) = 1.2V$



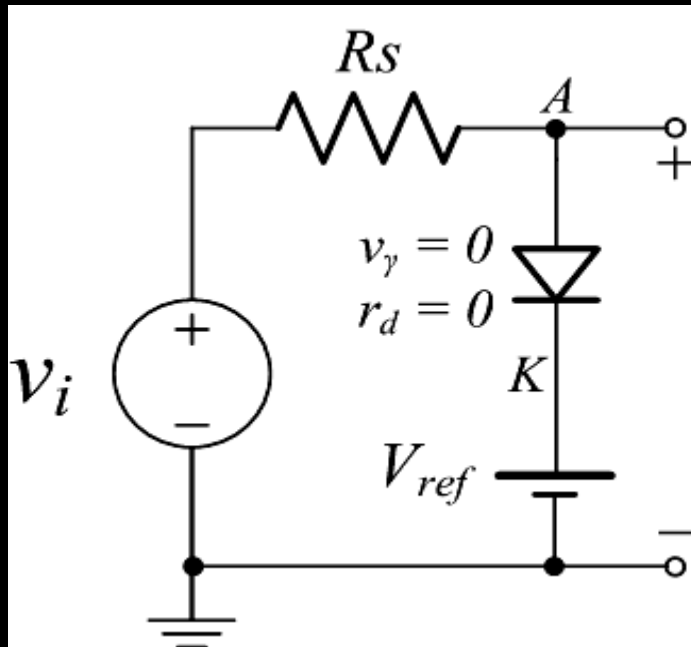
Comparison of Si, Ge & GaAs diode V_γ



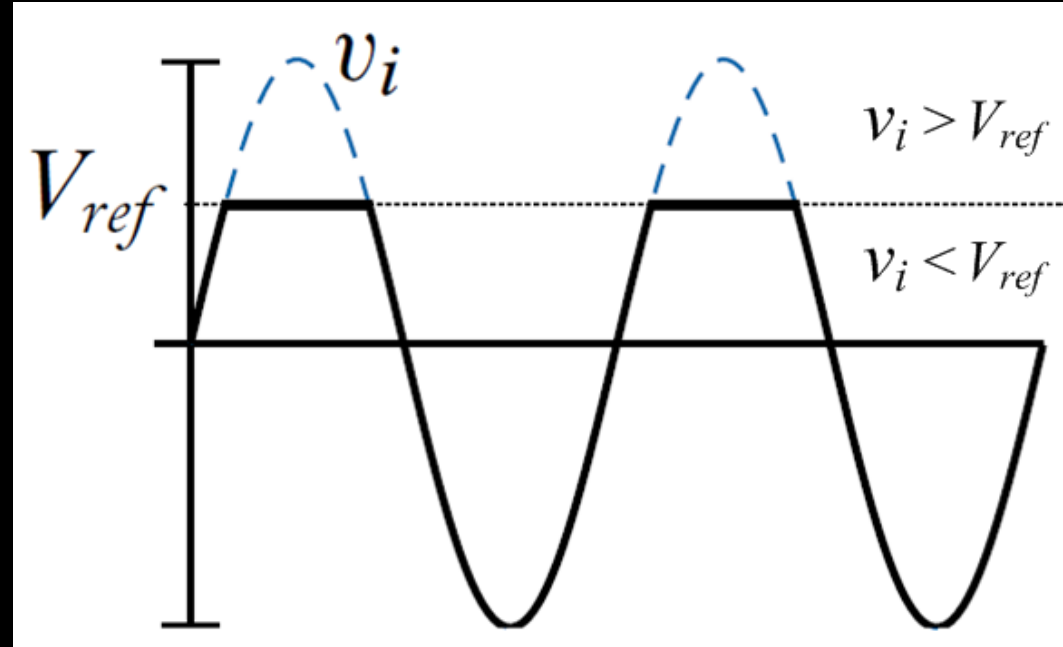
CLIPPER CIRCUITS

- ❑ Clippers are networks that employ diodes to “clip” away a portion of an input signal without distorting the remaining part of the applied waveform.
- ❑ In particular, when used with a DC bias voltage in series with the diode, the output signal can be limited to the reference voltage level of the DC voltage source.

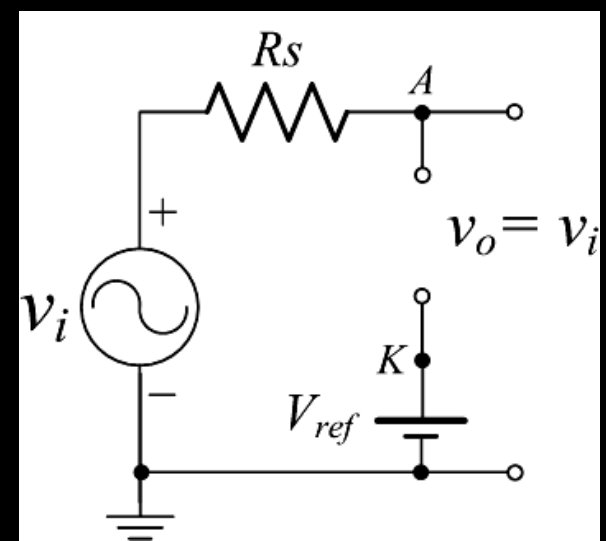
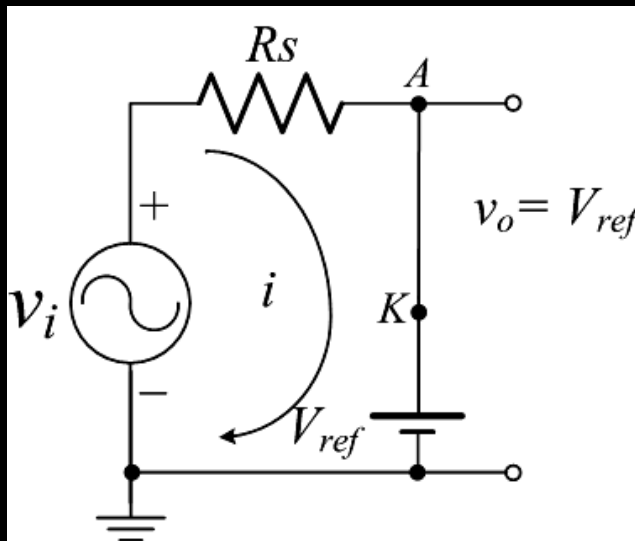
Biased Clipper with Ideal Diode



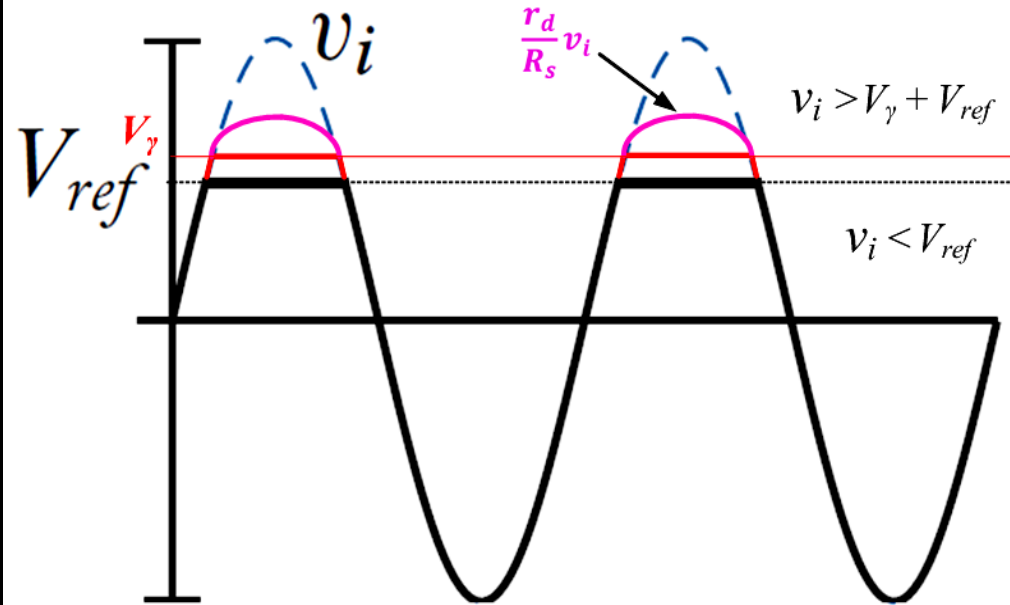
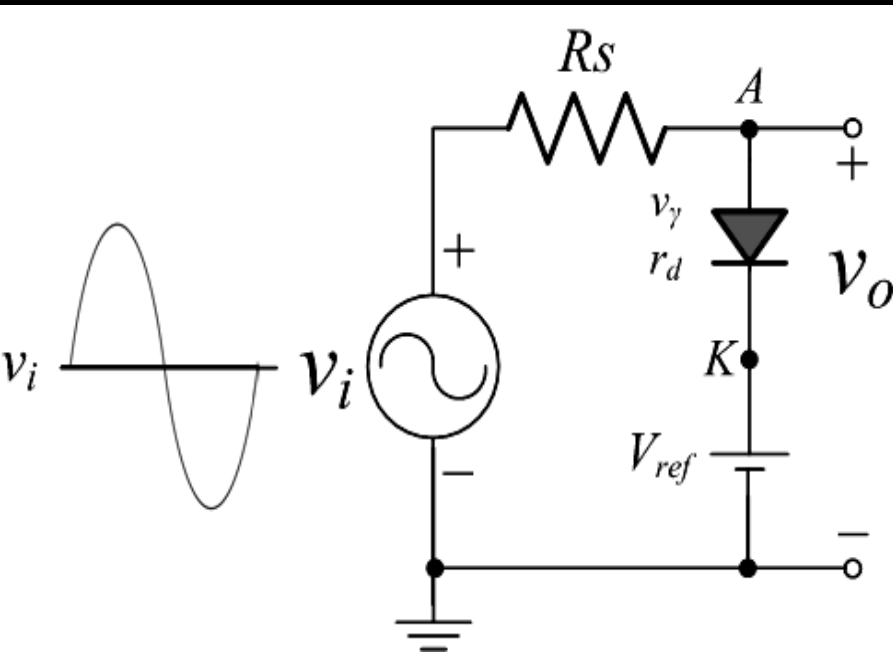
*Eqv. Ckt. : diode is forward bias
for $v_i \geq V_{ref}$.*



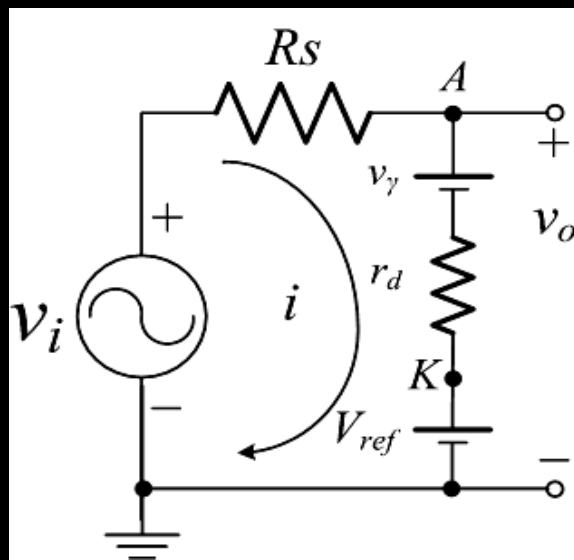
*Eqv. Ckt. : diode is reverse bias
for $v_i \leq V_{ref}$.*



Biased Clipper with Real Diode



Eqv. Ckt. : +ve half cycle of input, diode is forward bias. $v_i > v_\gamma + V_{ref}$

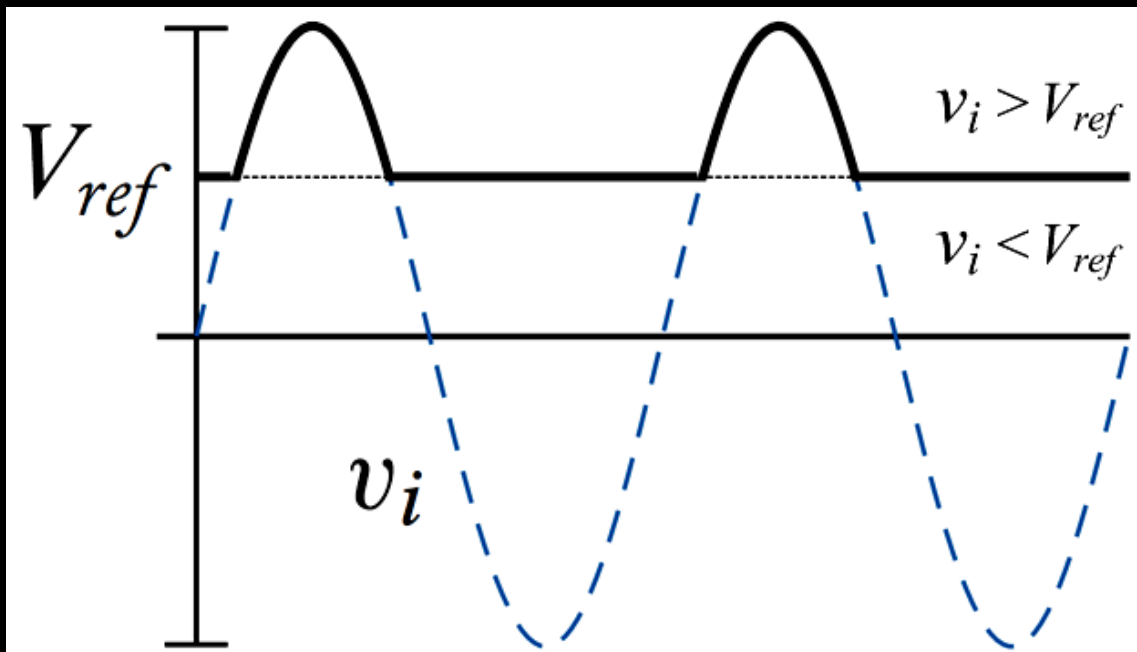
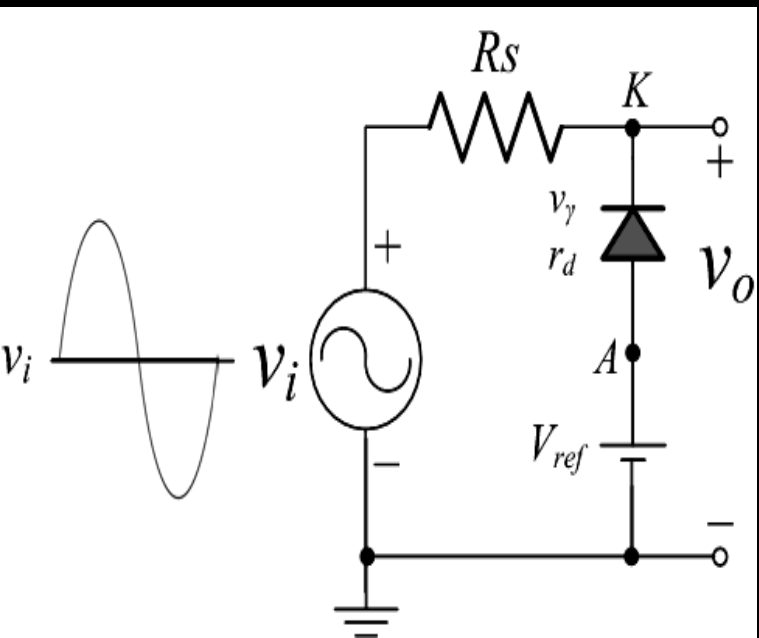


$$v_o = \frac{r_d}{R_s + r_d} v_i + \frac{R_s}{R_s + r_d} (v_\gamma + V_{ref})$$

usually $r_d \ll R_s$

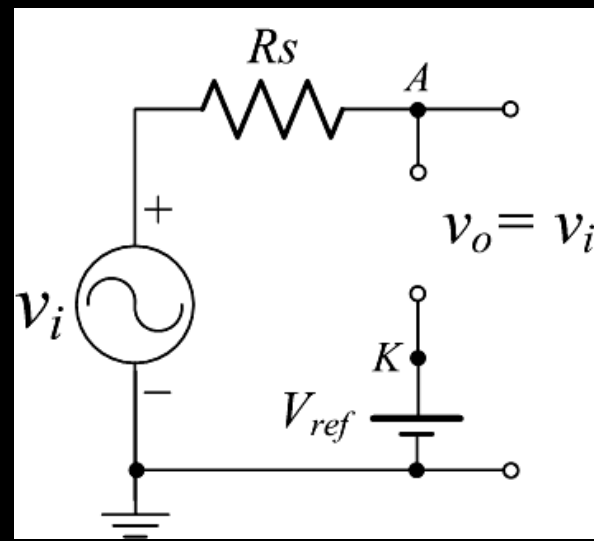
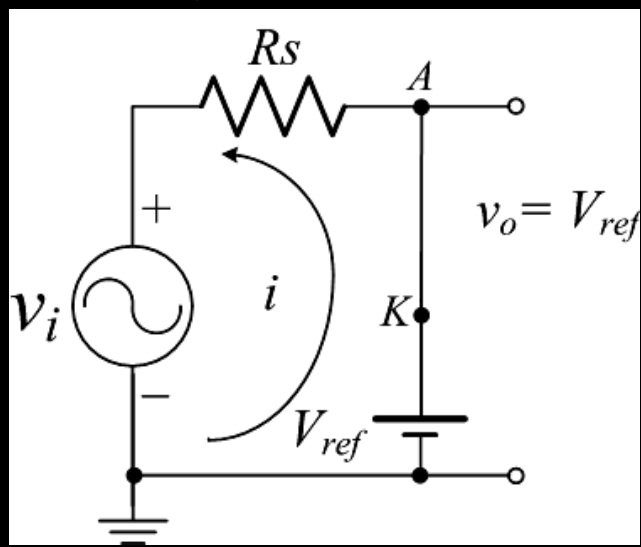
$$v_o \approx \frac{r_d}{R_s} v_i + (v_\gamma + V_{ref})$$

Biased Clipper



*Eqv. Ckt. : diode is forward bias
for $V_i \leq V_{ref}$.*

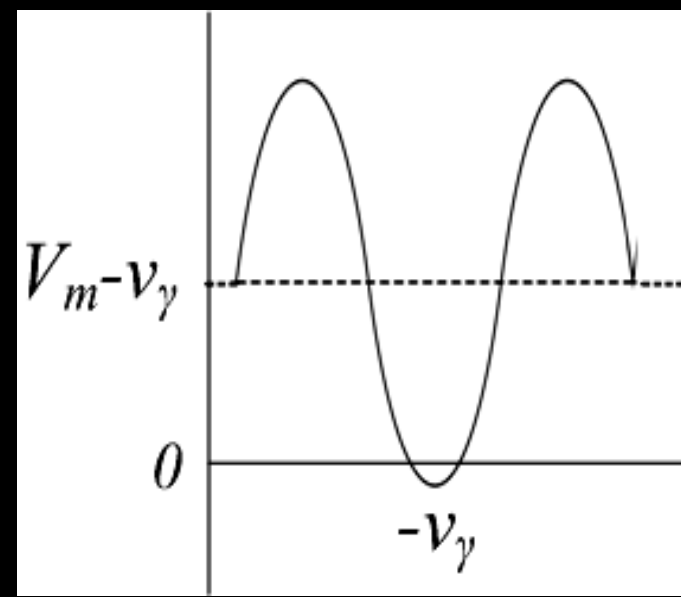
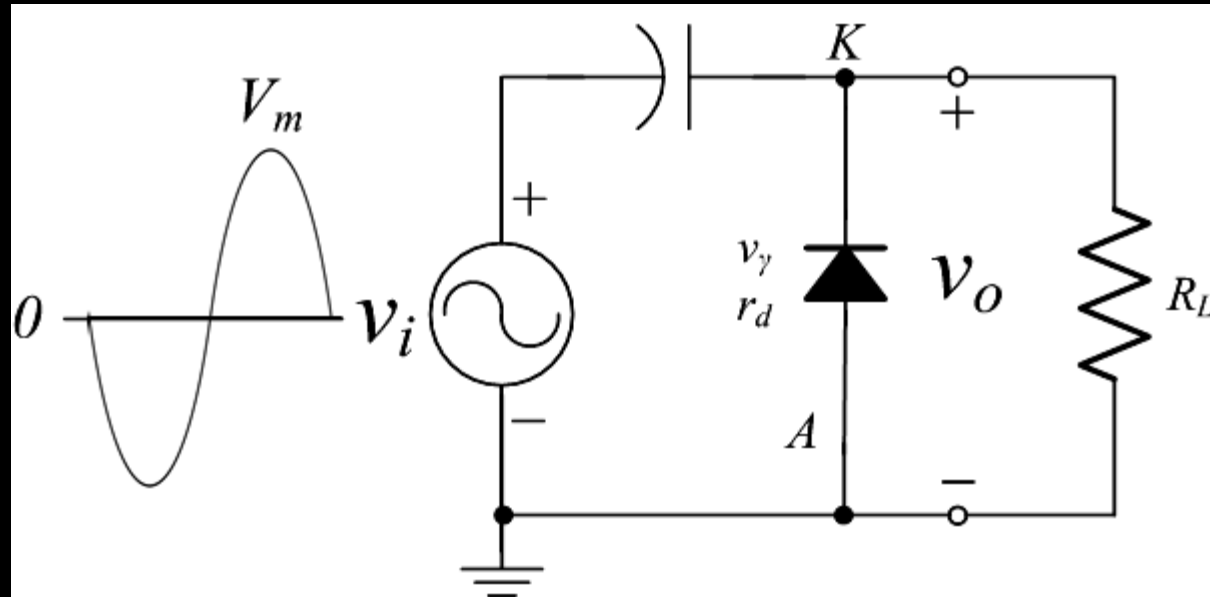
*Eqv. Ckt. : diode is reverse bias
for $V_i \geq V_{ref}$.*



DIODE CLAMPERS

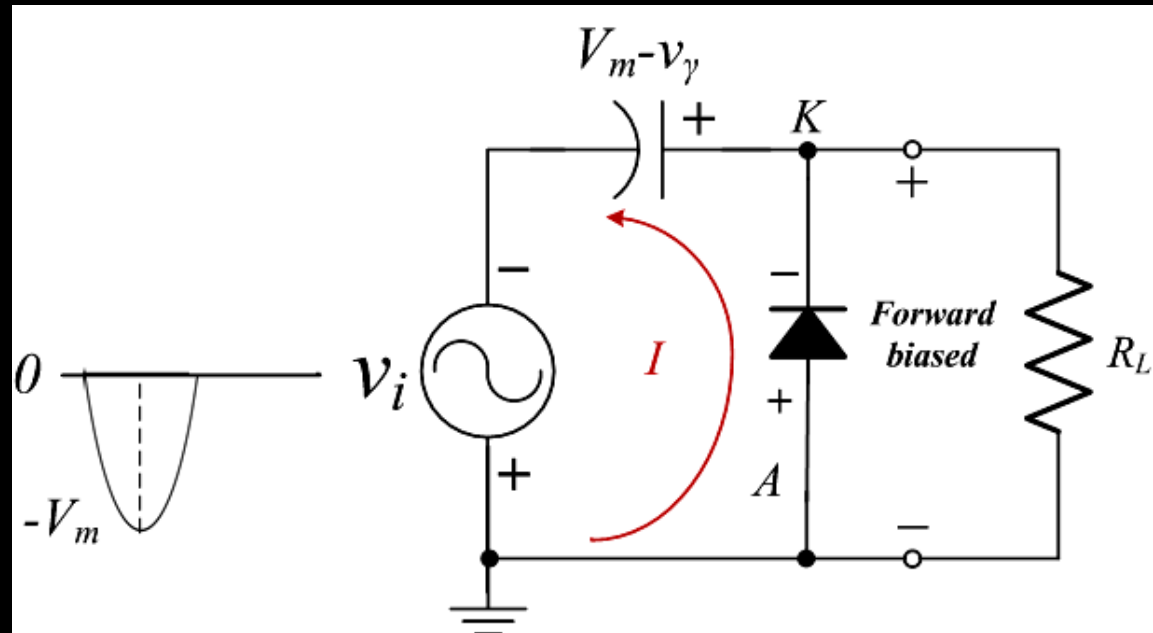
A Clamper circuit adds a DC level to an AC voltage. A Diode Clamper inserts a DC level to the input signal waveform.

POSITIVE DIODE CLAMPER

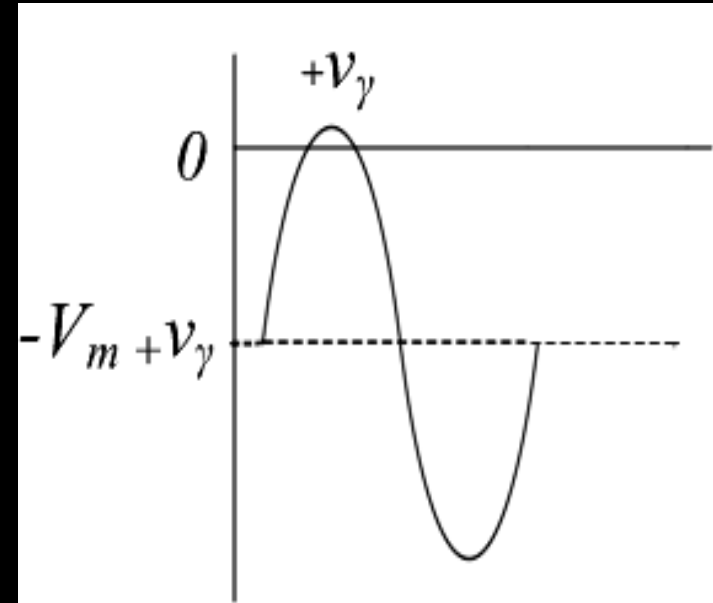
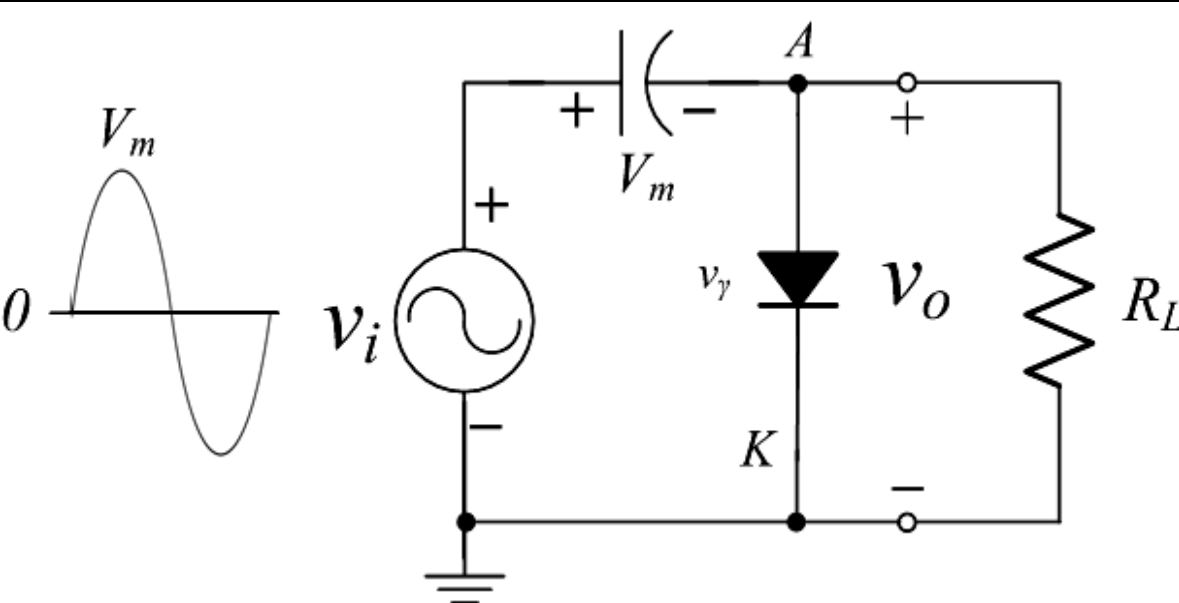


Capacitor charging process for the -ve half cycle of the input signal when Diode is ON.

When the signal crosses $-V_m$ of the -ve half cycle, Diode is OFF.

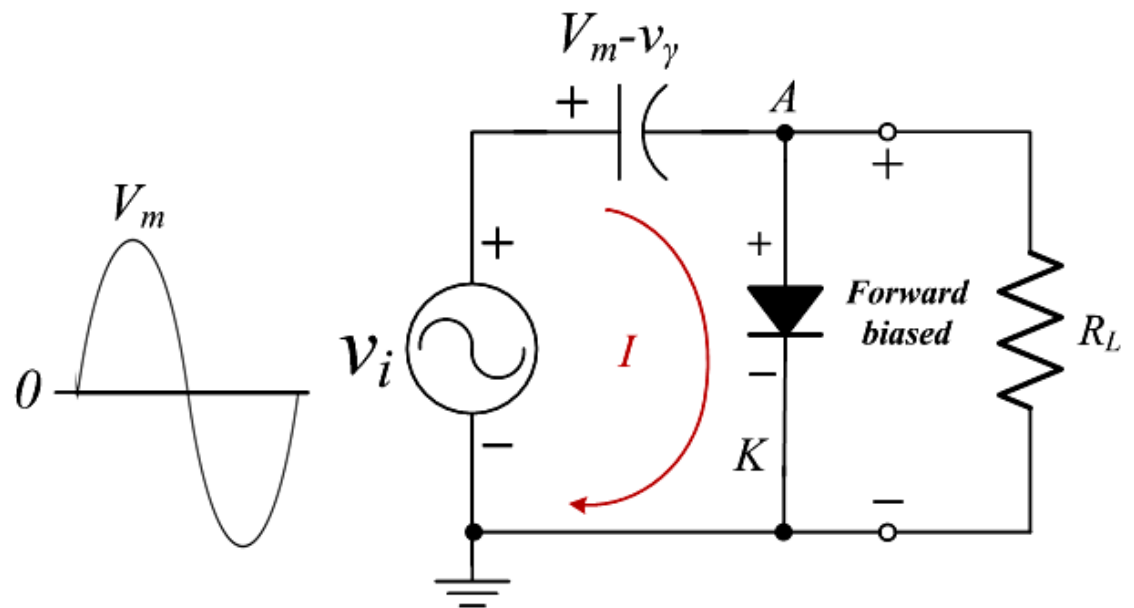


NEGATIVE DIODE CLAMPER



Capacitor charging process for the +ve half cycle of the input signal when Diode is ON.

When the signal crosses V_m of the +ve half cycle, Diode is OFF.

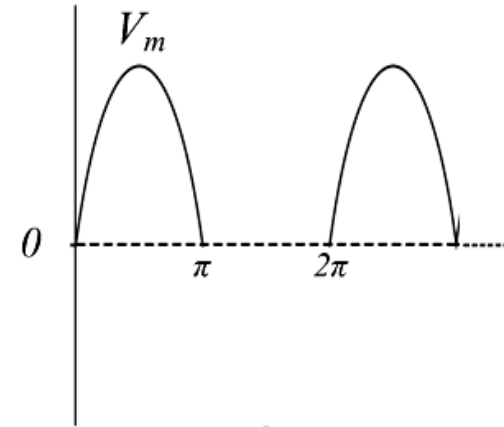
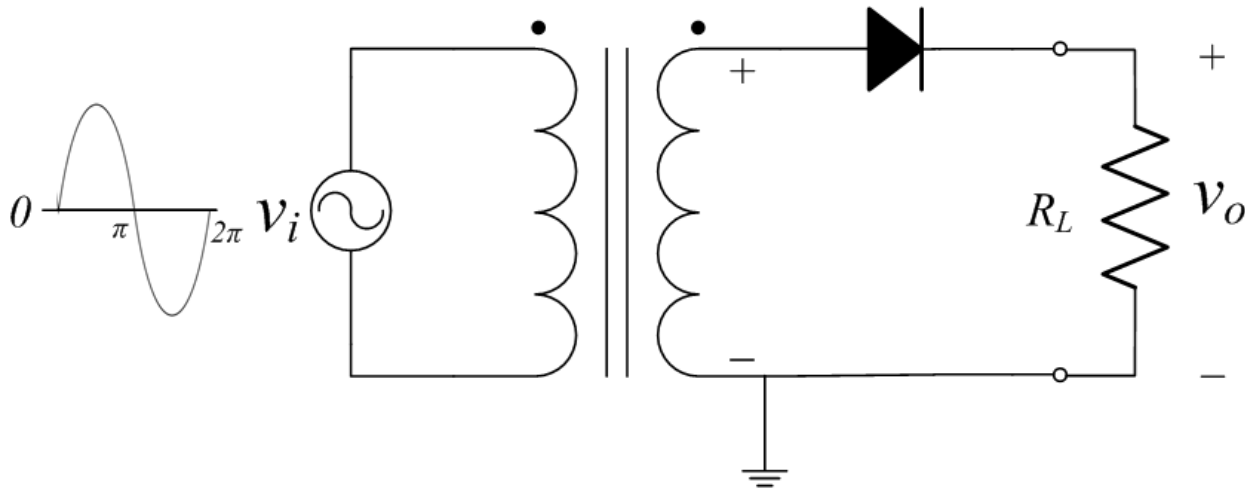


DIODE RECTIFIER

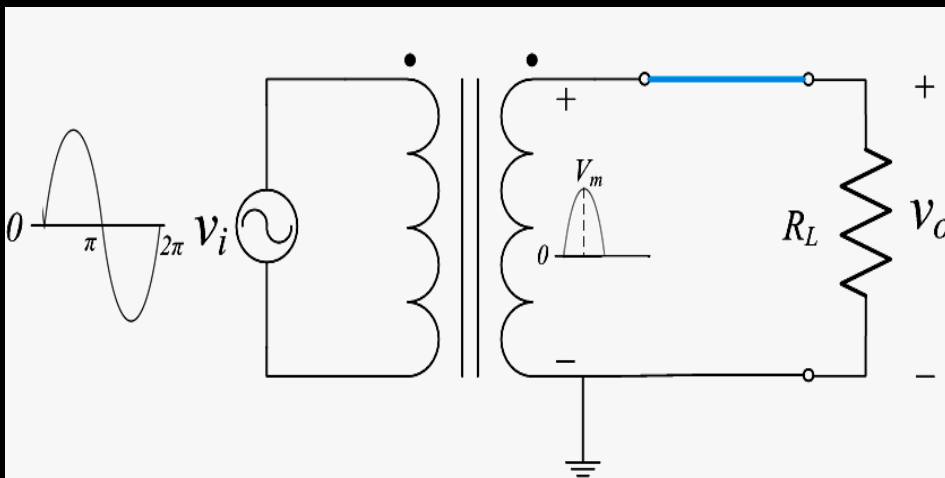
A Rectifier that converts an AC voltage to unidirectional voltage is used as DC power supply

Half Wave Rectifier

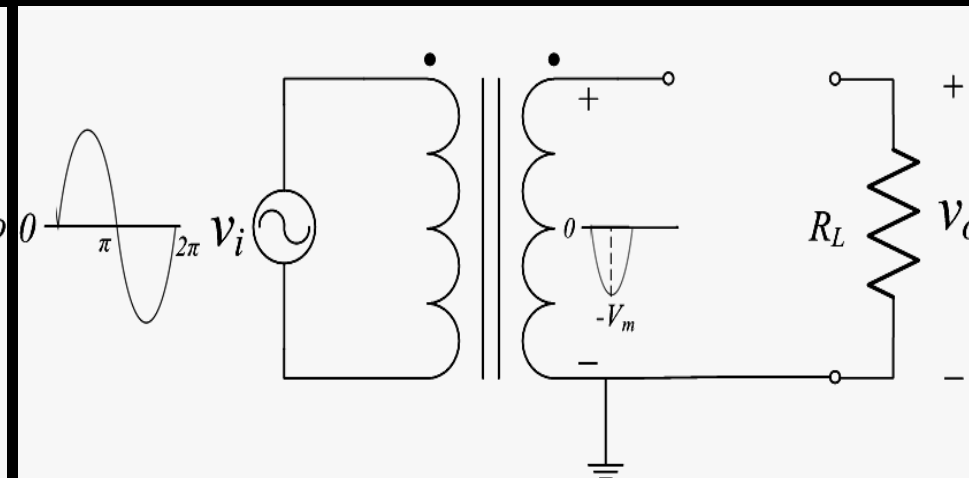
It only allows current from the positive half cycle of the input to flow through the load resistor



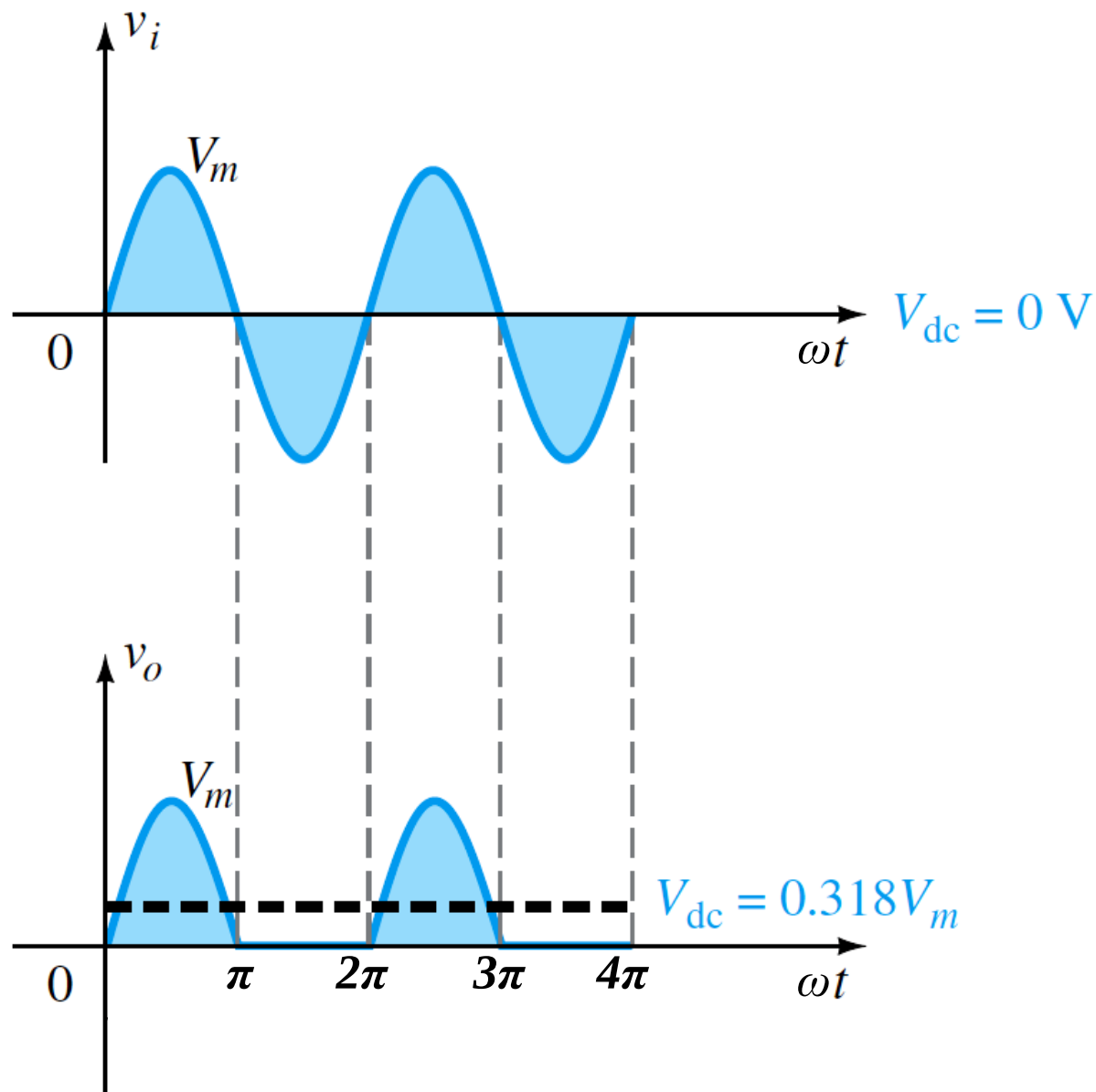
For +ve half cycle Diode is ON



For -ve half cycle Diode is OFF



$$v_i(\omega t) = V_m \sin \omega t$$



$$v_o(\omega t) = \begin{cases} V_m \sin \omega t, & \text{for } 0 \leq \omega t \leq \pi \\ 0 & , \text{for } \pi \leq \omega t \leq 2\pi \end{cases}$$

$$v_0(\omega t) = \begin{cases} V_m \sin \omega t, & \text{for } 0 \leq \omega t \leq \pi \\ 0 & , \text{ for } \pi \leq \omega t \leq 2\pi \end{cases}$$

DC Value :

$$V_{0(av)} = \frac{1}{2\pi} \int_0^{2\pi} v_0(\theta) d\theta = \frac{V_m}{\pi} = 0.318V_m$$

RMS Values :

$$V_{0(rms)} = \left[\frac{1}{2\pi} \int_0^\pi v_0^2 d(\omega t) \right]^{\frac{1}{2}} = \left[\frac{1}{2\pi} \int_0^\pi V_m^2 \sin^2 \omega t d(\omega t) \right]^{\frac{1}{2}}$$
$$= \frac{V_m}{2}$$

$$I_{0(rms)} = \frac{V_{0(rms)}}{R_L} = \frac{V_m}{2R_L}$$

Ripple Voltage :

$$v_r(t) = v_0(t) - V_{0(av)}$$

$$v_r^2(t) = v_0^2(t) - 2v_0(t)V_{0(av)} + V_{0(av)}^2$$

$$\overline{v_r^2(t)} = \overline{v_0^2(t)} - 2\overline{v_0(t)V_{0(av)}} + \overline{V_{0(av)}^2}$$

$$\overline{v_r^2(t)} = \overline{v_0^2(t)} - 2V_{0(av)}^2 + V_{0(av)}^2$$

$$V_{r(rms)}^2 = V_{0(rms)}^2 - V_{0(av)}^2$$

$$V_{r(rms)}^2 = V_{0(rms)}^2 - V_{0(av)}^2$$

RMS Value of Ripple Voltage :

$$V_{r(rms)} = \sqrt{V_{0(rms)}^2 - V_{0(av)}^2}$$

Ripple Factor Γ :

$$RF = \Gamma = \frac{V_{r(rms)}}{V_{0(av)}} = \sqrt{\left(\frac{V_{r(rms)}}{V_{0(av)}}\right)^2 - 1}$$

$$\Gamma = \sqrt{\left(\frac{V_m/2}{V_m/\pi}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2}\right)^2 - 1} = \mathbf{1.21 \text{ or } 121\%}$$

Efficiency η :

$$\eta = \frac{P_{ac}}{P_{dc}} = \frac{\frac{V_{dc}^2}{R_L}}{\frac{V_{rms}^2}{R_L}} = \frac{\left(\frac{V_m}{\pi}\right)^2}{\left(\frac{V_m}{2}\right)^2} = 0.406 \text{ or } 40.6\%$$

Form Factor :

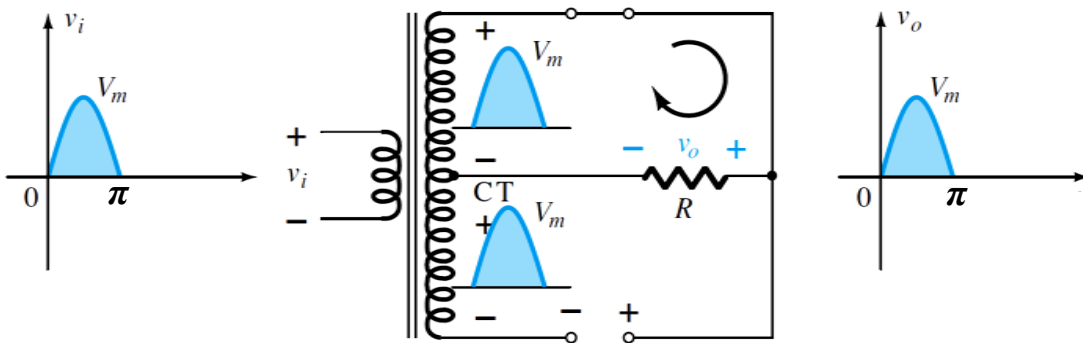
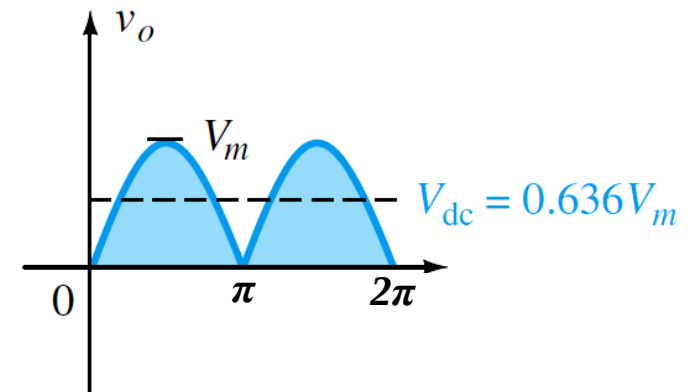
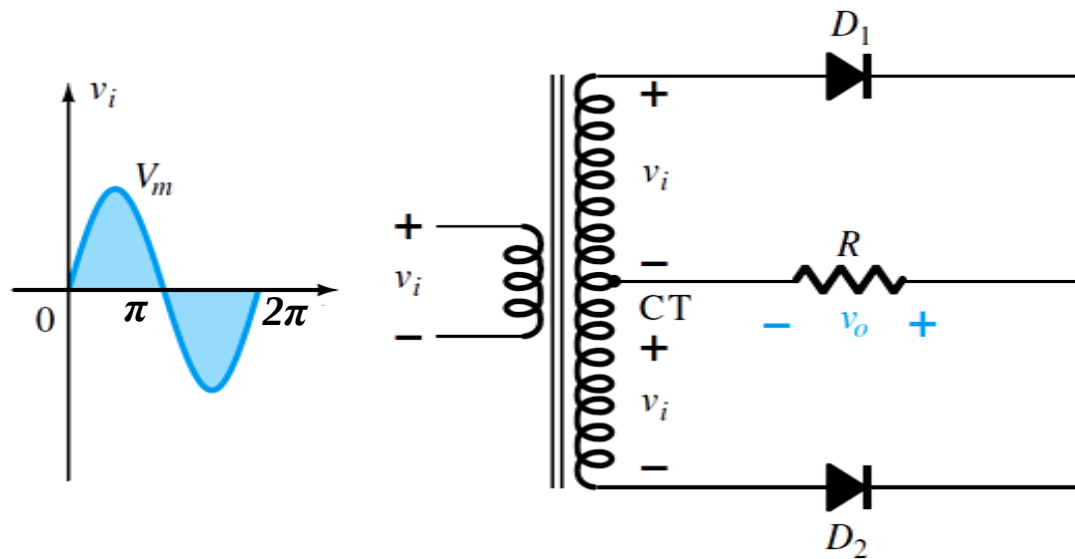
$$\text{Form Factor} = \frac{V_{rms}}{V_{dc}} = \frac{\frac{V_m}{2}}{\frac{V_m}{\pi}} = \frac{\pi}{2} = 1.57$$

Peak Inverse Voltage PIV :

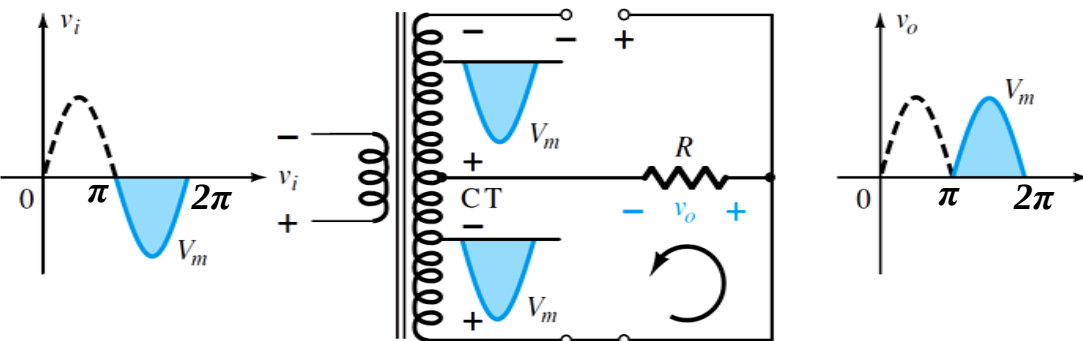
The maximum reverse voltage that a diode can withstand without destroying the junction. For half-wave rectifier, PIV is V_m .

Full Wave Rectifier

Center-Tapped Transformer

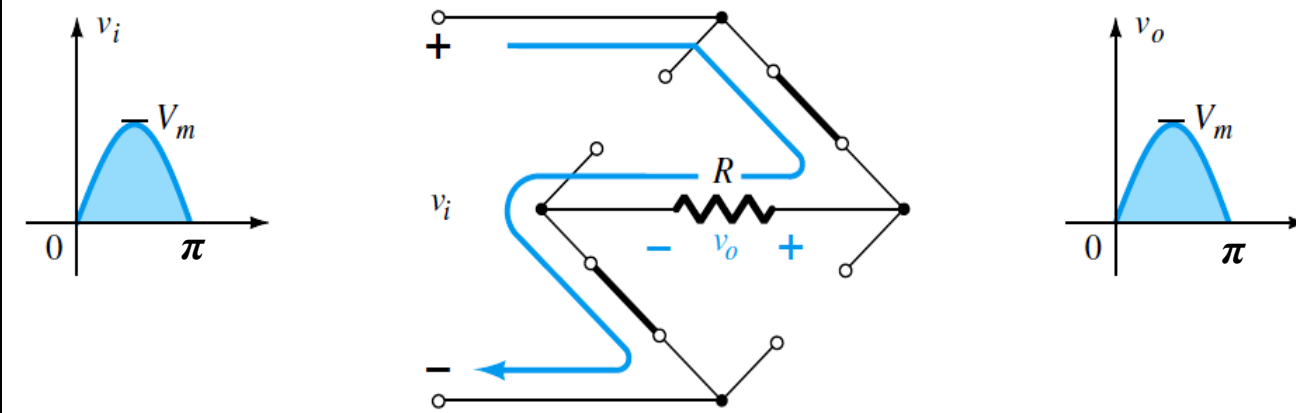
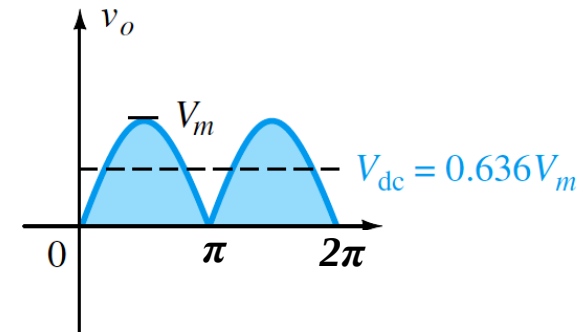
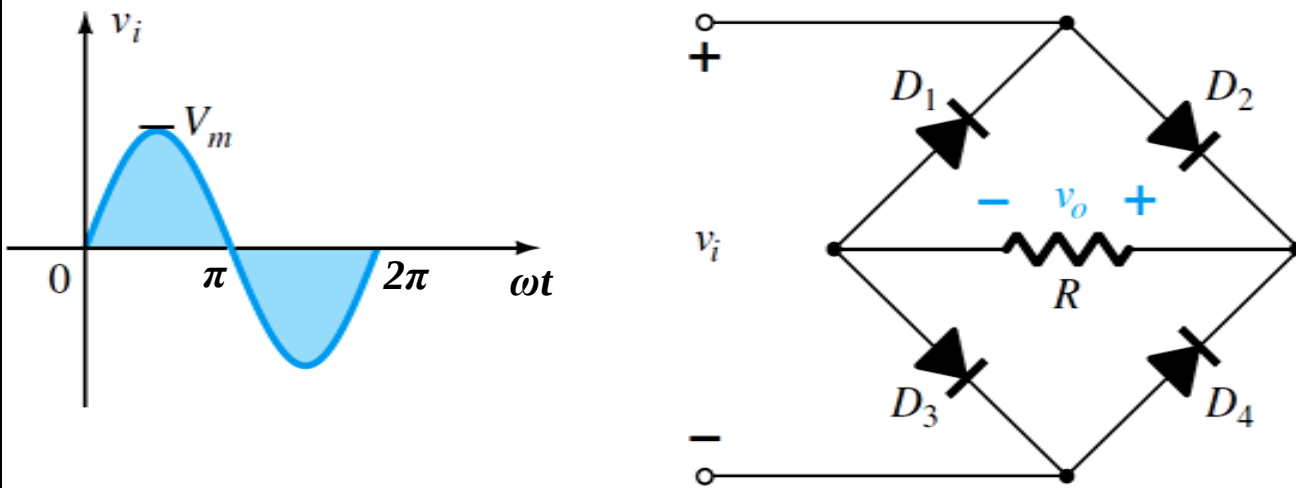


*For +ve half cycle
D1 ON & D2 OFF*

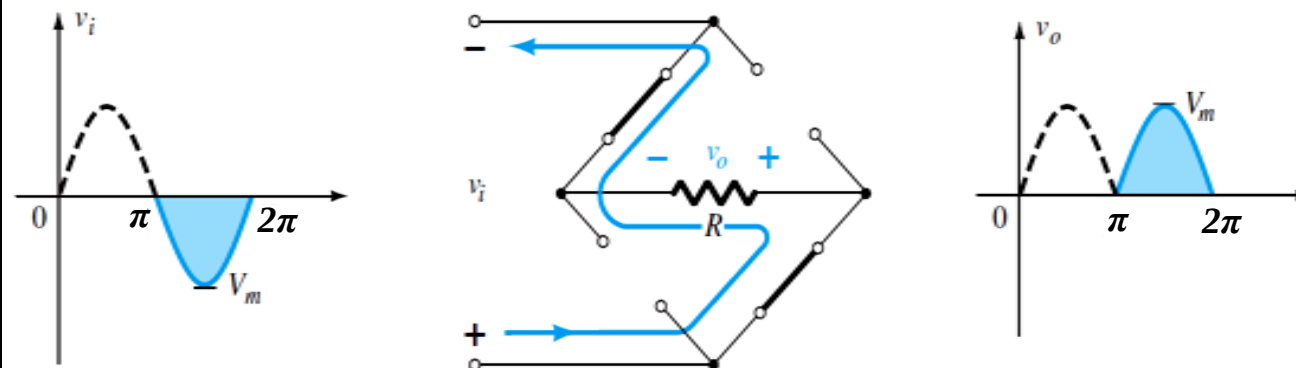
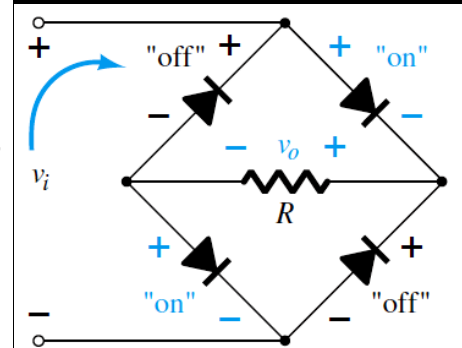


*For -ve half cycle
D2 ON & D1 OFF*

Full Wave Rectifier Bridge Network



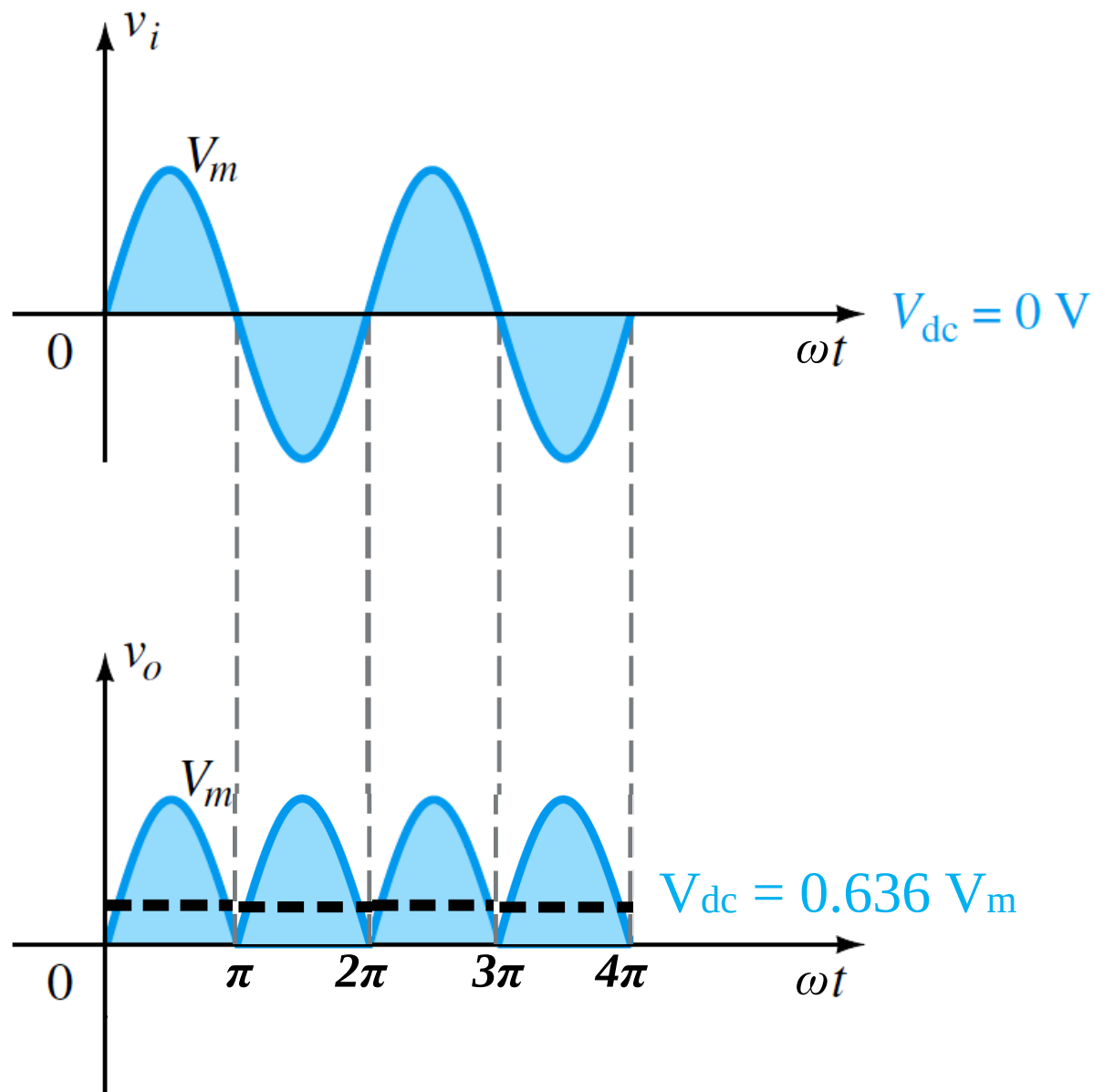
D_2 & D_3 ON



D_1 & D_4 ON

$$v_i(\omega t) = V_m \sin \omega t$$

$$= V_m \sin \theta$$



$$v_o(\omega t) = \begin{cases} V_m \sin \omega t, & \text{for } 0 \leq \omega t \leq \pi \\ -V_m \sin \omega t, & \text{for } \pi \leq \omega t \leq 2\pi \end{cases}$$

$$v_0(\omega t) = \begin{cases} V_m \sin \theta, & \text{for } 0 \leq \omega t \leq \pi \\ -V_m \sin \theta, & \text{for } \pi \leq \omega t \leq 2\pi \end{cases}$$

DC Value :

$$V_{0(\text{av})} = \frac{1}{2\pi} \int_0^{2\pi} v_0(\theta) d\theta = \frac{2V_m}{\pi} = 0.636V_m$$

RMS Values :

$$\begin{aligned} V_{0(rms)} &= \left[\frac{2}{2\pi} \int_0^\pi v_0^2 d(\omega t) \right]^{\frac{1}{2}} = \left[\frac{2}{2\pi} \int_0^\pi V_m^2 \sin^2 \omega t d(\omega t) \right]^{\frac{1}{2}} \\ &= \frac{V_m}{\sqrt{2}} = 0.707V_m \\ I_{0(rms)} &= \frac{V_{0(rms)}}{R_L} = \frac{V_m}{\sqrt{2}R_L} \end{aligned}$$

Ripple Voltage :

$$v_r(t) = v_0(t) - V_{0(av)}$$

$$V_{r(rms)}^2 = V_{0(rms)}^2 - V_{0(av)}^2$$

$$V_{r(rms)} = \sqrt{V_{0(rms)}^2 - V_{0(av)}^2}$$

$$V_{r(rms)}^2 = V_{0(rms)}^2 - V_{0(av)}^2$$

RMS Value of Ripple Voltage :

$$V_{r(rms)} = \sqrt{V_{0(rms)}^2 - V_{0(av)}^2}$$

Ripple Factor Γ :

$$RF = \Gamma = \frac{V_{r(rms)}}{V_{0(av)}} = \sqrt{\left(\frac{V_{0(rms)}}{V_{0(av)}}\right)^2 - 1}$$

$$\Gamma = \sqrt{\left(\frac{V_m/\sqrt{2}}{2V_m/\pi}\right)^2 - 1} = \sqrt{\frac{\pi^2}{8} - 1} = \mathbf{0.483 \text{ or } 48.3\%}$$

Efficiency η :

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{\frac{V_{dc}^2}{R_L}}{\frac{V_{rms}^2}{R_L}} = \frac{\left(\frac{2V_m}{\pi}\right)^2}{\left(\frac{V_m}{\sqrt{2}}\right)^2} = 0.812 \text{ or } 81.2\%$$

Form Factor :

$$\text{Form Factor} = \frac{V_{rms}}{V_{dc}} = \frac{\frac{V_m}{\sqrt{2}}}{\frac{2V_m}{\pi}} = \frac{\pi}{2\sqrt{2}} = 1.11$$

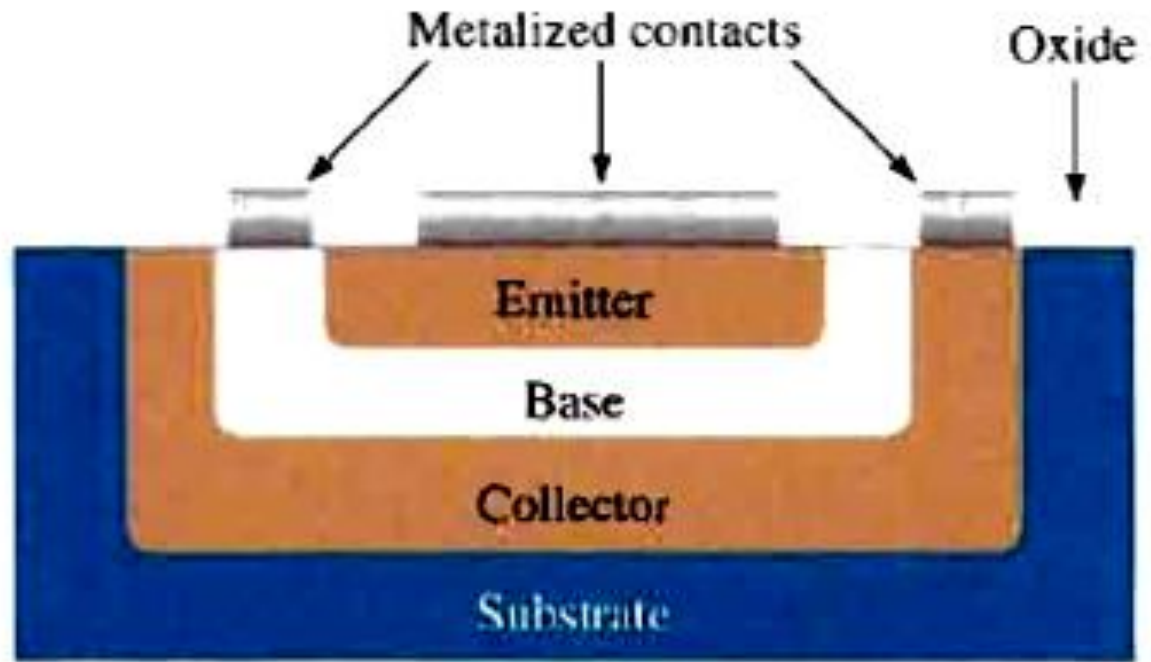
Peak Inverse Voltage PIV :

The maximum reverse voltage that a diode can withstand without destroying the junction. For half-wave rectifier, PIV is V_m .

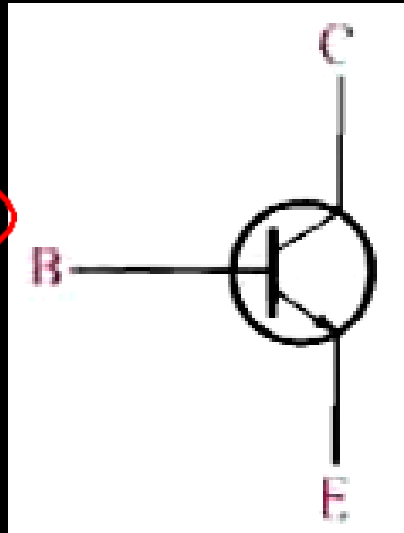
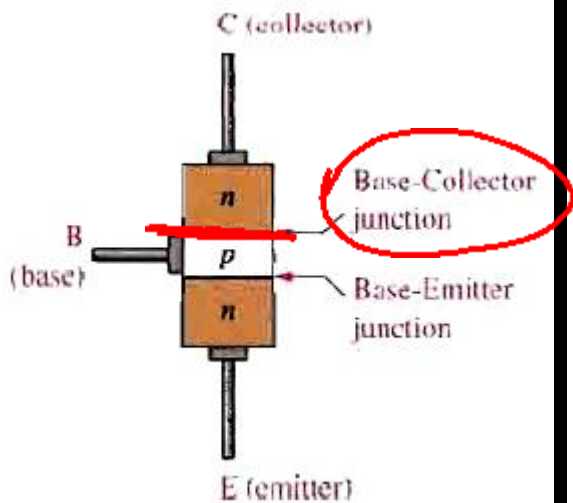
For Full-Wave Center-Tapped Transformer rectifier, PIV is $2V_m$.

For Full-Wave Bridge Network rectifier, PIV is V_m .

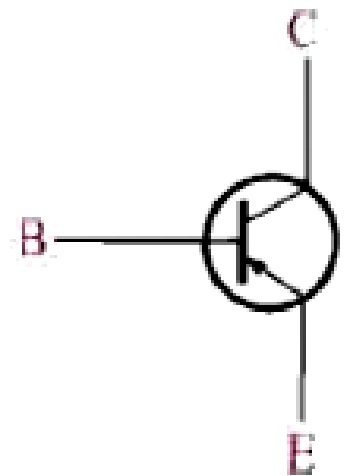
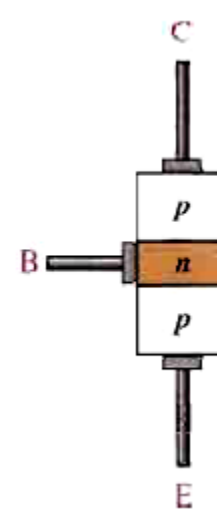
BIPOLAR JUNCTION TRANSISTOR (BJT)



npn-Transistor



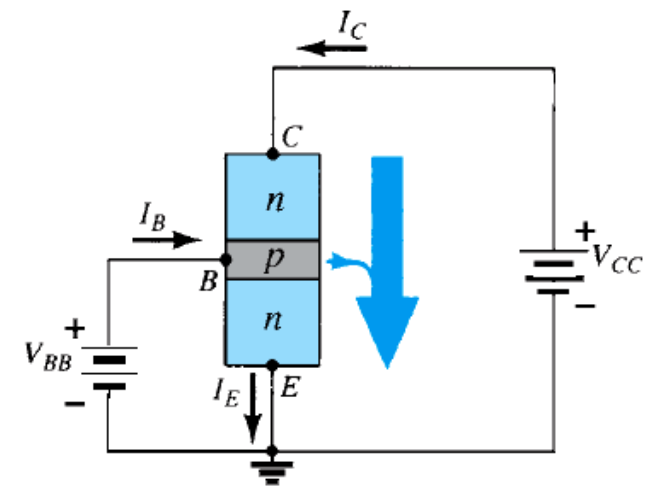
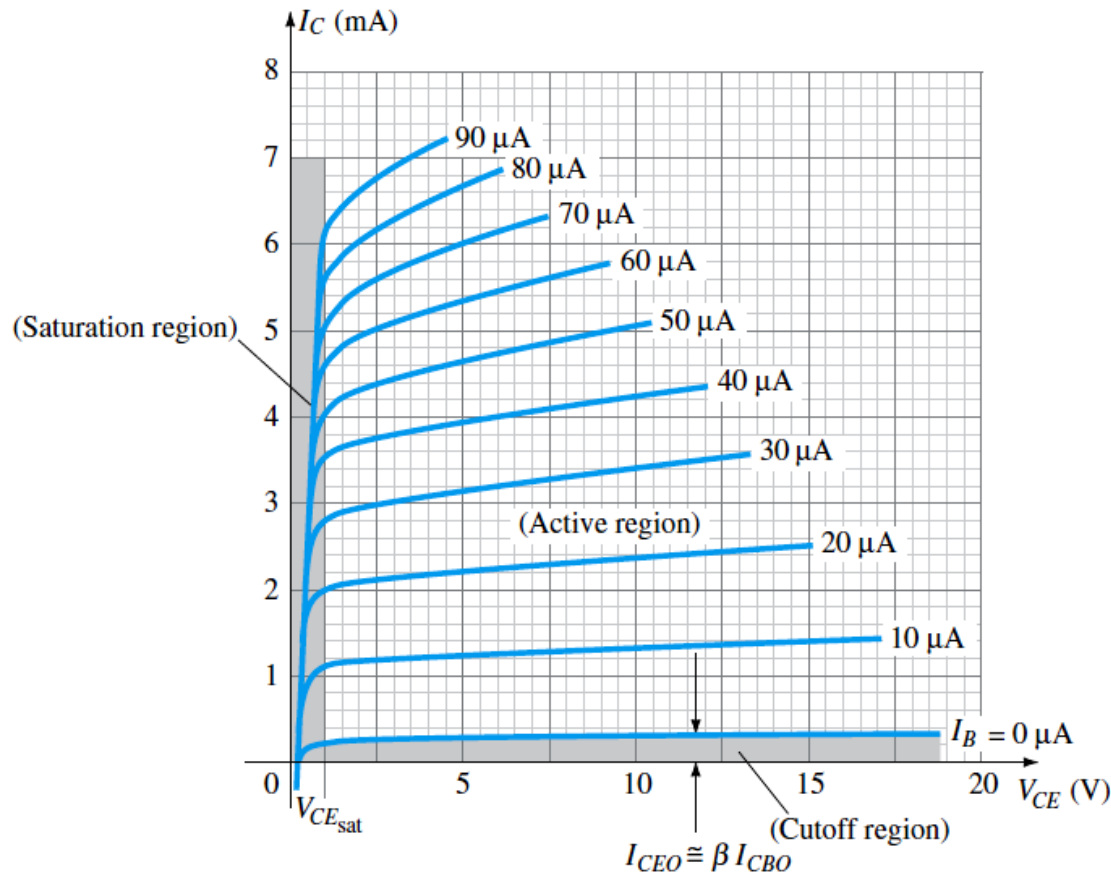
pnp-Transistor



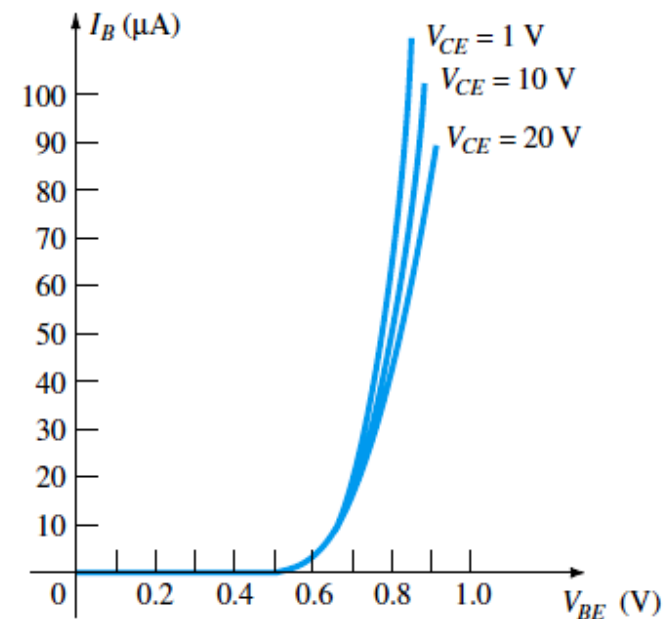
COMMON EMITTER CONFIGURATION

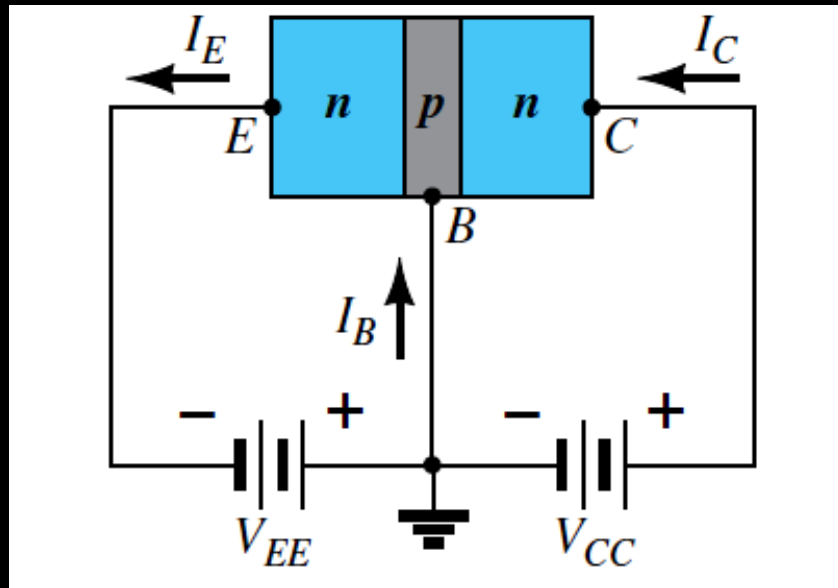
The emitter is common to both the input and output terminals (in this case common to both the base and collector terminals).

Collector Characteristics



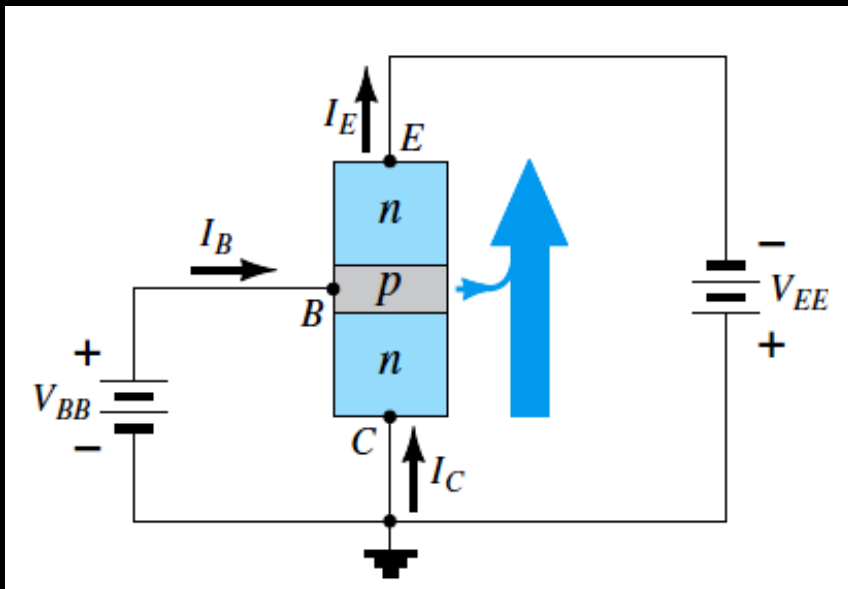
Base Characteristics





COMMON BASE CONFIGURATION

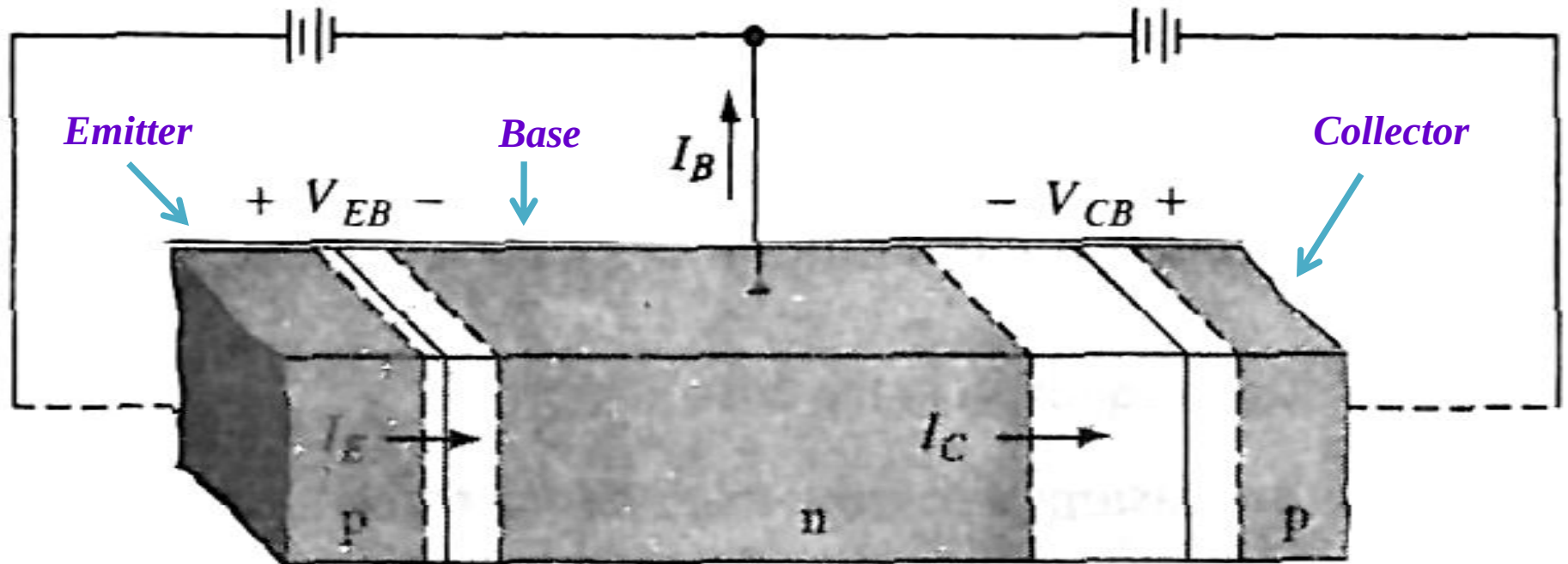
The base is common to both the input and output terminals (in this case common to both the emitter and collector terminals).



COMMON COLLECTOR CONFIGURATION

The collector is common to both the input and output terminals (in this case common to both the base and emitter terminals).

BJT - REGION OF OPERATION

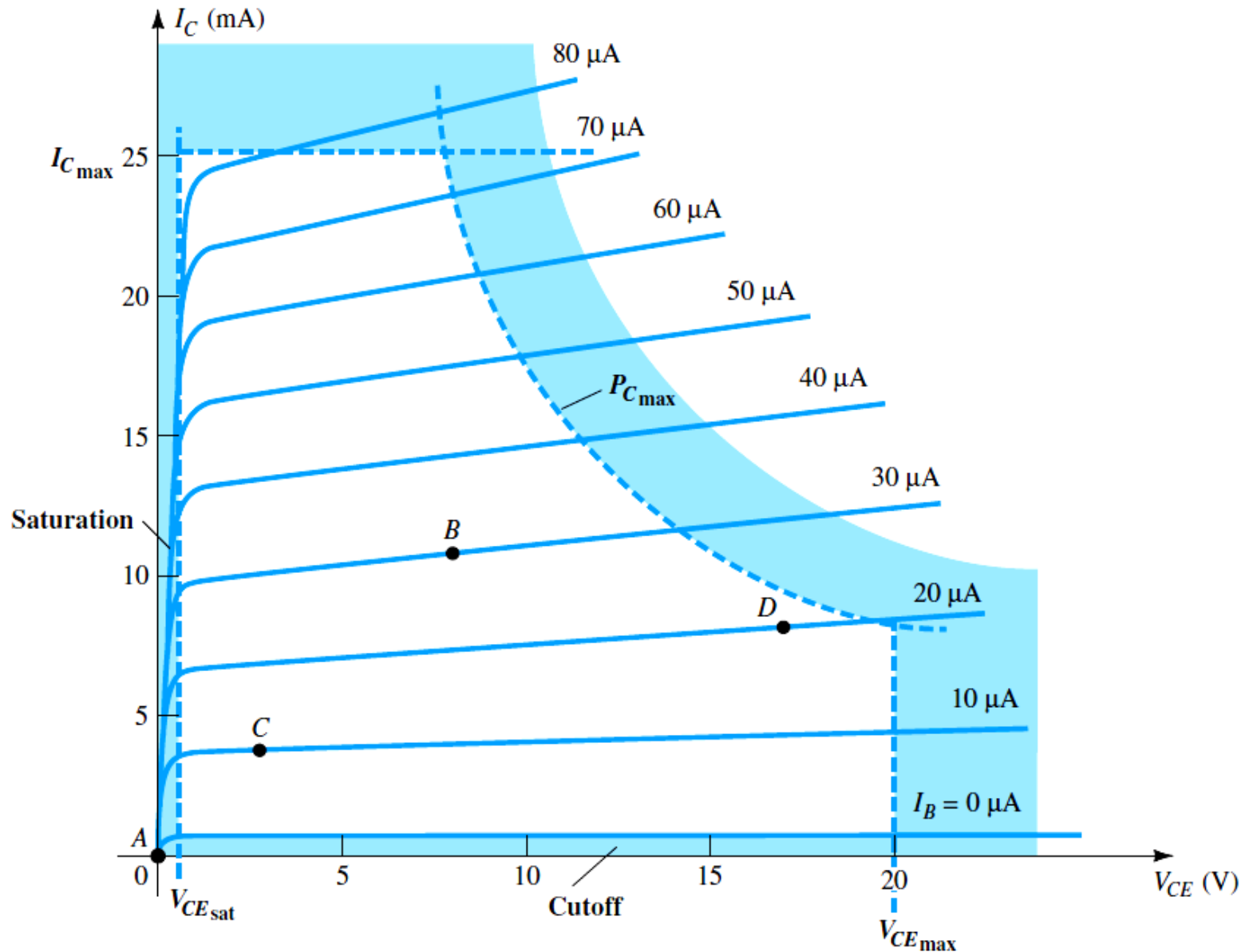


Active Region : EB junction forward bias & BC junction are reverse bias

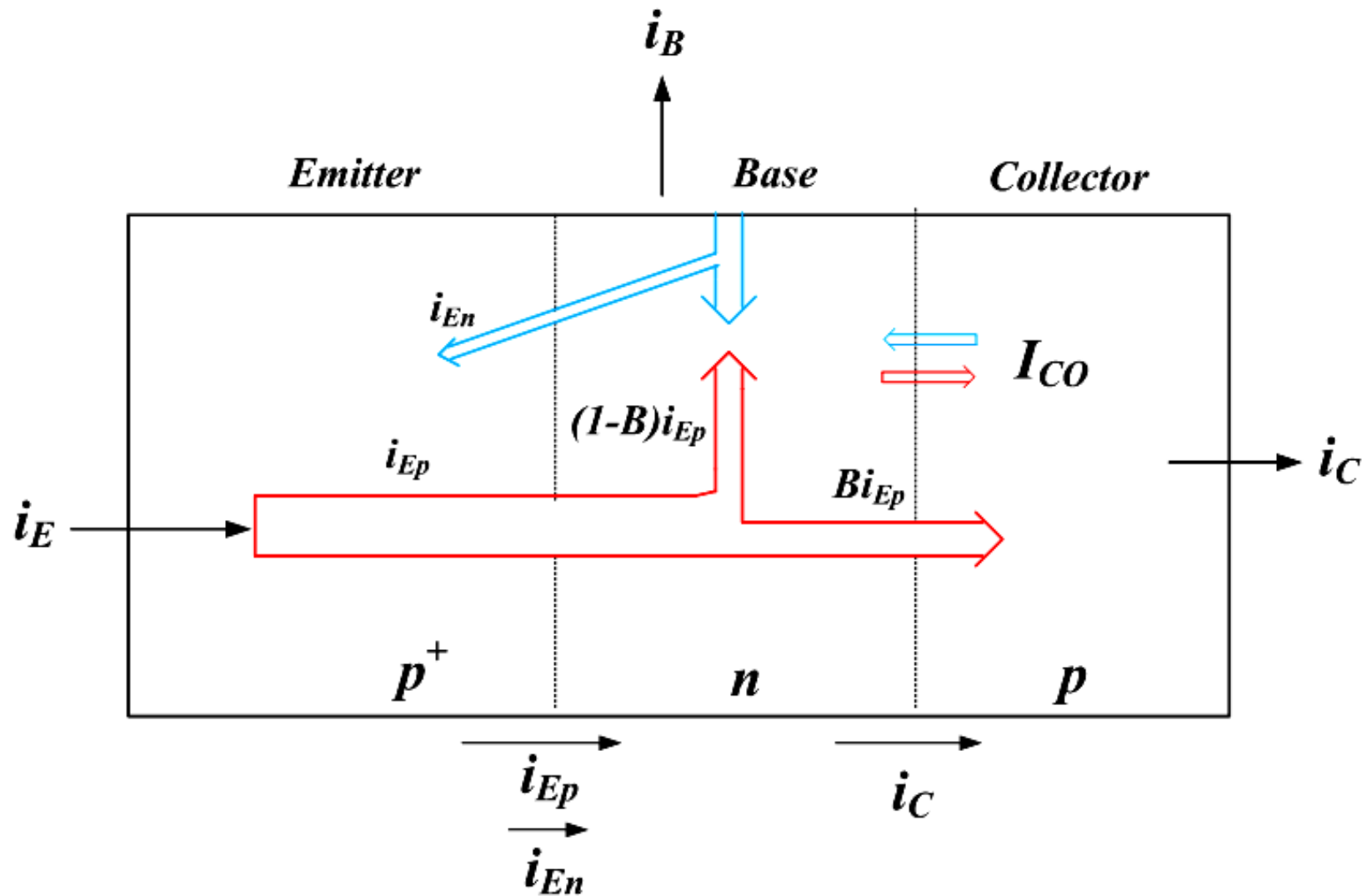
Cut-off Region : EB & BC junction are reverse bias

Saturation Region : EB & BC junction are forward bias

OPERATING REGIONS OF JUNCTION TRANSISTOR



FUNDAMENTALS OF BJT OPERATION



AMPLIFICATION WITH BJT

Emitter Current : $i_E = i_{En} + i_{Ep}$

Collector Current : $i_C = B i_{Ep}$

Base Transport Factor : B

Emitter Injection Efficiency : $\gamma = \frac{i_{Ep}}{i_{En} + i_{Ep}}$

For Efficient Transistor

$$\gamma \approx 1 \quad B \approx 1$$

The Ratio :

$$\frac{i_C}{i_E} = \frac{B i_{Ep}}{i_{En} + i_{Ep}} = B \gamma = \alpha$$

Current transfer ratio, which represents the emitter-to-collector current amplification. $\alpha < 1$

Base Current : $i_B = i_{En} + (1 - B) i_{Ep}$

The Ratio :

Hence

$$\frac{i_C}{i_B} = \frac{B i_{Ep}}{i_{En} + (1 - B) i_{Ep}}$$

$$\frac{i_C}{i_B} = \frac{B \frac{i_{Ep}}{(i_{En} + i_{Ep})}}{1 - B \frac{i_{Ep}}{(i_{En} + i_{Ep})}}$$

$$\frac{i_C}{i_B} = \frac{B \gamma}{1 - B \gamma} = \frac{\alpha}{1 - \alpha} = \beta$$

Base-to-Collector current amplification factor. Since α is near unity, it is clear that β can be large for a good transistor, and the collector current is large compared with the base current

$$\beta = \frac{\alpha}{1 - \alpha}$$

Hence

$$\alpha = \frac{\beta}{\beta + 1}$$

From

$$i_E = i_C + i_B$$

and

$$i_C = \beta i_B$$

$$i_E = \beta i_B + i_B$$

Hence

$$i_E = (\beta + 1) i_B$$

Considering Collector Reverse Saturation Current

$$i_C = \alpha i_E + I_{CO} \quad \text{or} \quad i_E = \frac{i_C - I_{CO}}{\alpha}$$

$$i_E = i_C + i_B$$

$$i_C = \frac{i_C - I_{CO}}{\alpha} - i_B$$

$$\alpha i_C = i_C - I_{CO} - \alpha i_B$$

Hence

$$i_C = \beta i_B + (\beta + 1) I_{CO}$$

$$i_C = \frac{\alpha}{1 - \alpha} i_B + \frac{1}{1 - \alpha} I_{CO}$$